THE FUEL — FULL MIND MAP

Chemistry · Chapter 7.F · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Fuel Basics, Fossil Fuels & Calorific Value

- Petroleum natural fuel hai, jabki coke, tar aur coal gas ko processing/destructive distillation se obtain kiya ja sakta hai. Coal, petroleum aur natural gas fossil fuels hain; inme natural gas comparatively cleaner fossil fuel maana jata hai because combustion par coal/diesel ke comparison me pollutants aur carbon emissions kam hote hain. [Q1–Q3]
- Calorific value ka matlab complete combustion par released heat per unit mass hota hai. Common fuels me Hydrogen ka calorific value sabse high hota hai (source approx. 150,000 kJ/kg), isliye high-energy applications/rocket-fuel context me important hai. [Q4]

2. Coal, Coke, Furnace Oil & Pollution

- India ke coal-based thermal plants ke baare me “koi seawater use nahi karta / water-stressed district me nahi hai / private ownership nahi hai” — teeno statements false hain. Coastal plants seawater cooling use kar sakte hain, water-scarce districts me plants hain aur private coal power producers bhi operate karte hain. [Q5]
- Coal ash me arsenic, lead, mercury jaise heavy metals ho sakte hain; coal-fired plants SO_2, NOx aur particulate pollutants emit karte hain, aur Indian coal ka ash content generally high hota hai. Copper smelting me slag heavy-metal leaching aur SO_2 major concerns hain. Indoor burning of coal harmful chemical pollution ka strong source hai. [Q6–Q8]
- Blast furnace me coke fuel + reducing agent dono ka role nibhata hai; silica removal mainly limestone-derived flux karta hai. Furnace oil crude-oil refining ka heavy residual product hai, industries/power generation me burn hota hai aur sulphur content ke karan SOx emissions produce kar sakta hai. [Q9–Q10]

3. Octane–Cetane Number & Diesel Engine

- Plastic-waste derived fuel ko source high-octane/lead-free “green fuel” batata hai. Octane number petrol ki anti-knock quality measure karta hai: higher octane = compression/knocking resistance better. Petrol quality commonly Octane number se express ki jati hai. [Q11–Q13]
- Cetane number diesel fuel ki ignition quality ka parameter hai; higher cetane generally shorter ignition delay aur better compression-ignition behaviour show karta hai. Diesel engine me air ko high compression se heat kiya jata hai aur phir diesel inject hokar ignite hota hai; carburetor petrol engines se associated hai. [Q14–Q15]

⚠ EXAM TRAP — Q15 — “vapour of diesel and air” ko literal mechanism mat samjho

► Source option (A) ko key diya gaya hai, lekin diesel engine me pehle sirf air compress hoti hai; diesel fuel baad me high-pressure injection se hot compressed air me spray hota hai.
► Exam option ko source context me yaad rakho, conceptual mechanism = compression ignition + fuel injection.

4. Biogas / Gobar Gas

- Biogas/Gobar gas ka chief combustible component Methane (CH_4) hai; mixture me commonly methane ke saath CO_2 hota hai aur traces me H_2S/moisture etc. ho sakte hain. Organic waste/cattle dung ka anaerobic fermentation/digestion biogas production ka basic process hai. [Q16–Q22]
- Source ke according cattle dung se methane-producing pilot gas plant S.V. Desai se linked hai; option me “C.B. Desai” typo diya gaya hai. Rice fields anaerobic conditions ki wajah se methane emit karte hain. [Q23–Q24]

⚠ EXAM TRAP — Q16 & Q23

► Q16 me “Cooking gas” ke options me Methane best fit hai because explanation biogas/gobar gas context use karti hai; lekin ordinary domestic cylinder cooking gas = LPG, mainly propane + butane.
► Q23 option “C.B. Desai” source typo hai; explanation S.V. Desai batati hai.

5. LPG: Composition, Storage & Safety

- LPG (Liquefied Petroleum Gas) mainly propane (C_3H_8) aur butane (C_4H_{10}) ka mixture hai; small amounts of other light hydrocarbons ho sakte hain. Domestic Indane/LPG context me butane + propane key combination hai. [Q25–Q30]

- LPG naturally almost odourless hoti hai, isliye leak detection ke liye Ethyl Mercaptan jaise strong-smelling odorant add kiya jata hai. Fuel-gas matching: CNG → methane-rich; LPG → propane/butane; coal gas → H_2 + CH_4 + CO; water gas → CO + H_2. [Q31–Q32]
- LPG ka main component methane nahi hai; methane pipeline natural-gas/PNG fuel ka principal component ho sakta hai. LPG cylinder me fuel pressure ke under liquid phase me stored hota hai; jab tak liquid present hai pressure mainly vapour pressure ke around rehta hai, isliye simple pressure gauge remaining quantity accurately show nahi karta. [Q33–Q35]

6. Natural Gas, CNG & LNG

- Natural gas ka dominant hydrocarbon Methane hai. CNG = Compressed Natural Gas; natural gas ko high pressure par compress karke transport/use kiya jata hai aur CNG me methane major constituent hota hai. [Q36–Q41]
- LNG = Liquefied Natural Gas; natural gas ko roughly –162°C tak cool karke liquid banaya jata hai, storage/transport easy hota hai. India ka first LNG terminal Dahej, Gujarat se associated hai. NGL (Natural Gas Liquids), not "NGM", me ethane, propane, butane aur natural gasoline jaise fractions aa sakte hain. [Q42]

⚠ EXAM TRAP — Q42 — LNG "high pressure" wording

► Question statement LNG ko extremely cold temperature + high pressure se liquefied batata hai, lekin standard LNG storage near-atmospheric pressure aur cryogenic temperature (~–162°C) par hota hai.
► Source explanation khud near-atmospheric pressure likhti hai; isliye statement wording technically inconsistent hai.

7. Fuel Blends, Engines & Emission Control

- Kerosene hydrocarbon mixture hai, isliye oxygen absent maana jata hai. Gasohol = gasoline/petrol + ethanol blend; geothermal energy non-conventional renewable source hai aur gobar gas methane-rich hoti hai. [Q43–Q46]
- Tetraethyl lead (TEL) historically petrol ka anti-knock additive tha, lekin lead toxicity ke karan largely phased out ho chuka hai. Automobile antifreeze me ethylene glycol commonly use hota hai. Heavy vehicles me diesel high torque + fuel efficiency ke karan useful hai; carburetor traditional petrol engines me, diesel me fuel injection system use hota hai. [Q47–Q51]
- Catalytic converters exhaust pollutants ko less harmful products me convert karte hain aur commonly platinum, palladium, rhodium catalysts use karte hain. Radial tyres, streamlined body aur multipoint fuel injection fuel economy improve karte hain; catalytic converter ka primary role emission control hai. [Q52–Q53]

⚠ EXAM TRAP — Q53 — Official key vs concept

► Source official key option (D) = 1, 3, 4 deta hai.
► Conceptually streamlined body (2) drag reduce karke fuel efficiency improve karti hai, jabki catalytic converter (4) ka primary purpose emissions control hai. Is question ko official-key specific trap ki tarah yaad rakho.

8. SAF, Hydrogen Fuel & Petroleum Products

- Sustainable Aviation Fuel (SAF) ke feedstocks me agricultural residues, corn grain, wastewater-treatment sludge/wet wastes aur wood-mill waste sab use ho sakte hain. SAF conventional jet fuel ke comparison me lower lifecycle carbon footprint target karta hai. [Q54]
- Hydrogen Fuel Cell Electric Vehicles me fuel cell hydrogen ki chemical energy ko electricity me convert karta hai; tailpipe se mainly water vapour + warm air nikalte hain. Hydrogen storage ke liye metal hydrides hydrogen absorb/release kar sakte hain. [Q55–Q56]
- Petroleum refining se asphalt/bitumen heavy by-product/fraction milta hai jo road construction binder me use hota hai. Source-era "Hydrocarbon Vision 2025" ko petroleum-product storage aur long-term energy security/self-reliance context se associate karta hai. [Q57–Q58]

✓ Source-question mapping preserved: Q1–Q58